Meme Coin Analysis Dashboard

Data Engineering Project Proposal

much data. such pipeline.

IDSC501 / OCT 2026



TRUMP, in two moods

what you want

Highest daily close $45.50

after you buy

Last supplied close $1.88



Daily closes, 18 Jan 2025–7 Oct 2026. Humorous captions describe no actual trade.

Market data and community activity



Price has a timestamp. So does the conversation.

Different formats
Different coverage
Different observation times

Our task: connect the evidence.

Project promotional image provides context, not evidence of social activity or investment merit.

Research questions

01 **How does observed activity relate to returns, volume and volatility?**

02 **What changes when coverage, reposts and collection delays enter the picture?**

The engineering question

Can another team replay the same inputs?

The price dataset is ready

14,773
daily records

10
coins

Last observation: 7 Oct 2026


DOGE


SHIB


PEPE


M


PENGU



TRUMP


SPX


BONK


FLOKI



BIAN

Fixed cohort. Common window: 4 Oct 2025–7 Oct 2026. This is not historical top-ten membership.

Collection scope

Source	Starting scope	Readiness
CMC CSV	10 coins, daily OHLCV	Available input
X via FxTwitter	5 selected coin accounts	Timeline checks succeeded
Reddit via Arctic Shift	3 candidate communities	Disabled pending access conditions

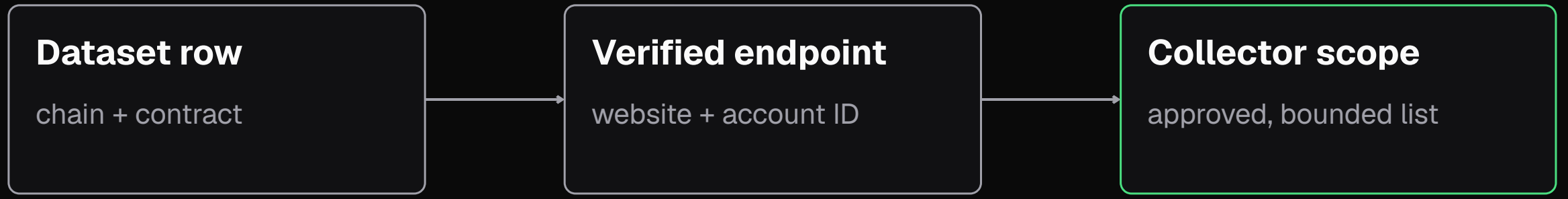
DOGE SHIB PEPE BONK SPX

Source checks: 14 Oct 2026. Selected accounts do not represent all X activity.

MemeChain supplies the starting links

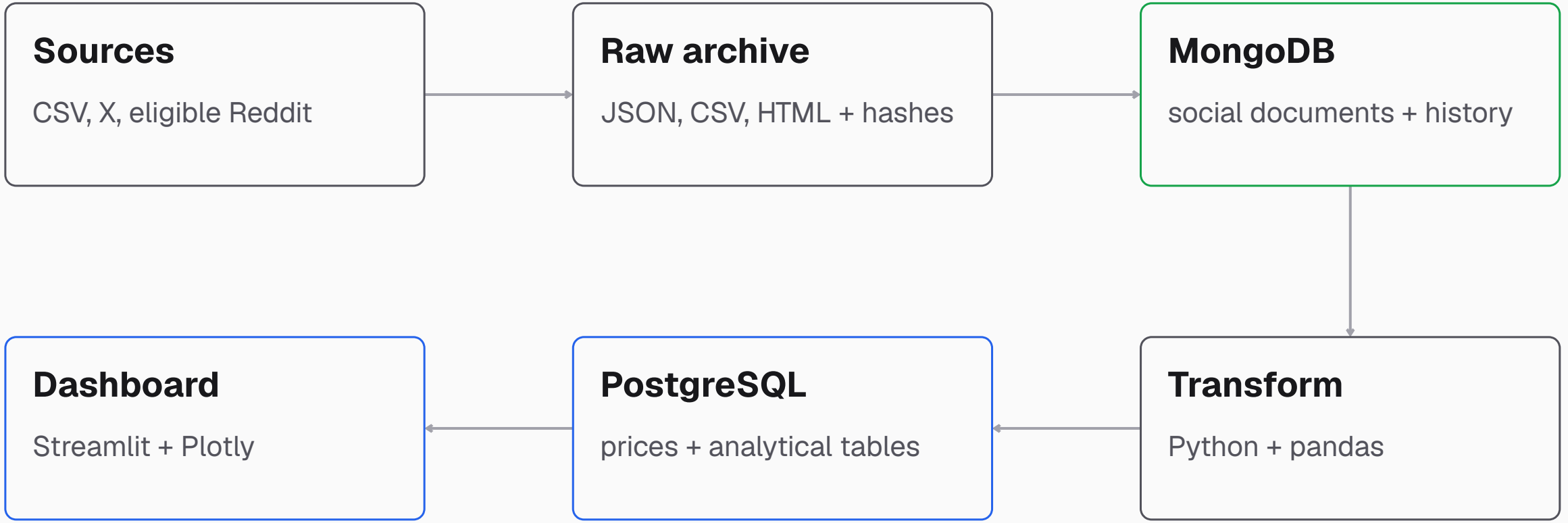
31,811

rows in the inspected social metadata table



Links are seeds. Verify identity, URL type and current access before collecting posts.

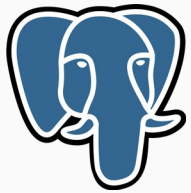
Source-to-dashboard architecture



Validated CSV prices enter PostgreSQL directly.

Charts read a published SQL build. Post details use stable IDs to retrieve MongoDB documents.

Why PostgreSQL + MongoDB?



PostgreSQL

Relationships and analysis

Prices, asset identities and SQL joins



Social documents

Platform fields and observation history

PostgreSQL also supports JSON.

We choose two models to separate responsibilities and demonstrate document-to-relational ETL.

Trade-off: two services, coordinated backups and cross-store consistency. Images are optional.

Preprocessing contracts

Same ticker ≠ same asset

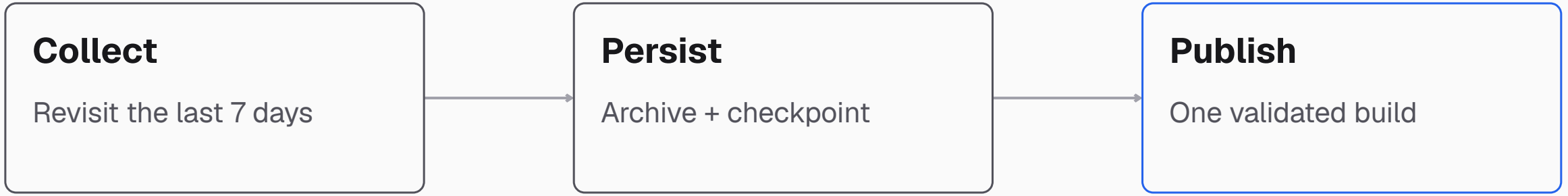
Contract	Rule
Asset identity	Verified chain + contract or native ID
Post identity	platform:item_type:native_id
Event time	UTC creation time, separate from fetch time
Text processing	Deduplicate, detect language, preserve context

Keep source zeros and unusual values with quality flags. Preserve raw responses for replay.

Daily batches, replayable inputs

02:00 UTC

11:00 KST



Monthly reconciliation

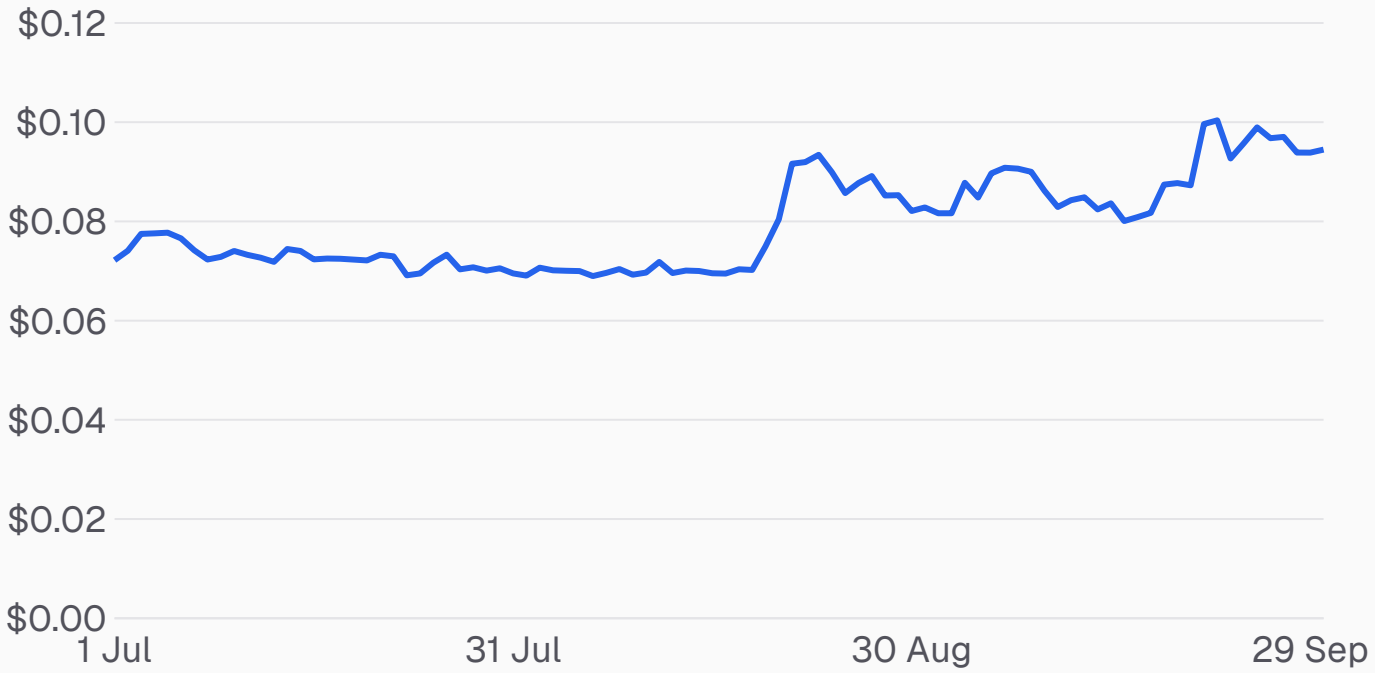
Last successful build survives a failed job.

Price refresh requires new CSV deliveries or restoration of the team collector.

Dashboard concept

Market overview

DOGE / 2026 Q3



Social activity

Awaiting the first eligible batch

Coverage: unavailable

Compare time series Explore posts Inspect data health

Concept layout. DOGE prices are real supplied data. The social panel contains no simulated observations.

Missing data has a meaning

NULL ≠ 0

0

A checked scope with no observed posts

NULL

A failed, partial or unavailable scope

“The API is down” is not a sentiment signal.

Evaluation targets

Correctness

Same inputs, same outputs
Recovery without duplicate records

200

items for manual review,
or the full eligible sample if smaller

< 2 s

SQL chart query p95 target

±7 days

Exploratory lead–lag range
At least 30 paired days

Targets, not achieved results. Report hardware, actual data volume and coverage.

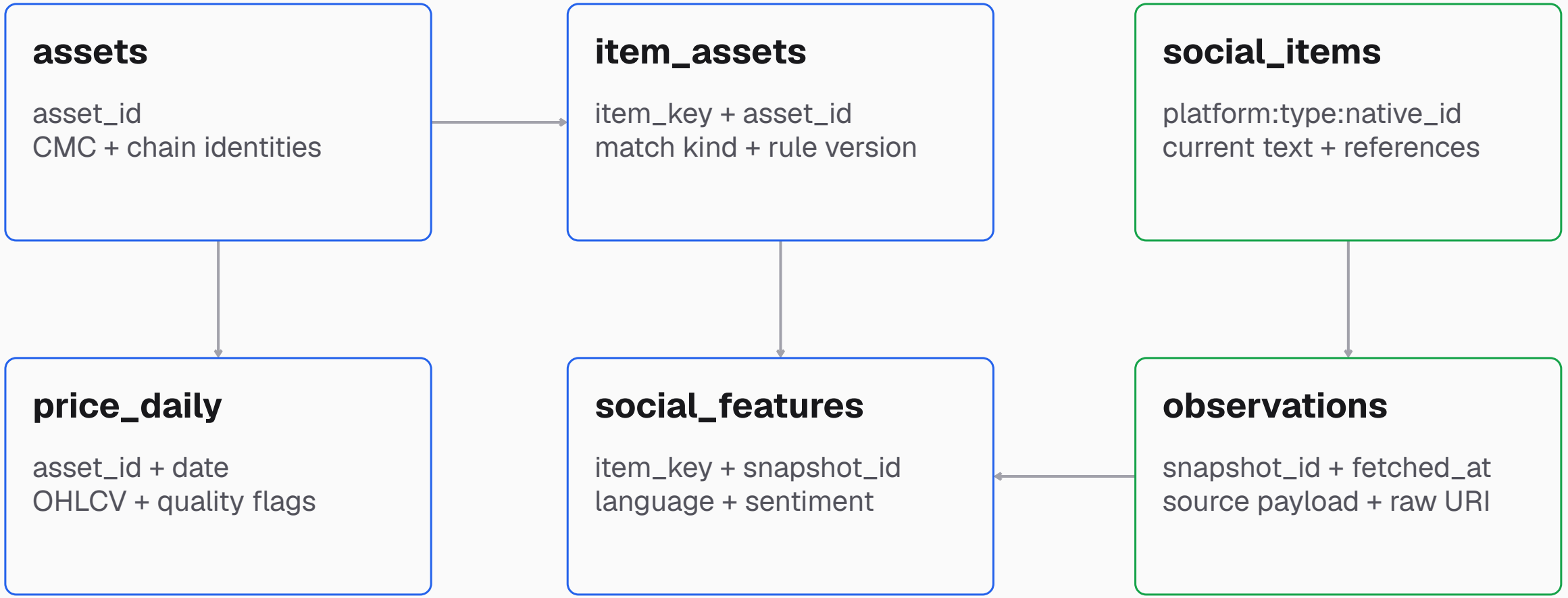
Six-week work plan

Week	Deliverable
01	Source validation and data contracts
02	Collectors and raw archives
03	Preprocessing and SQL/NoSQL ETL
04	Dashboard and provenance navigation
05	Quality, recovery and performance checks
06	Reproducible demo and final report

A dashboard with evidence behind every chart

Planning assumption. Team assignments and the official submission deadline remain TBD.

Appendix: storage ownership



Blue: PostgreSQL. Green: MongoDB. Validate cross-store references in ETL.

Appendix: limits and fallbacks

Dependency	Limit	Response
Price updates	Collector script missing	Import new validated CSVs
FxTwitter	Partial history or downtime	Expose coverage and freshness
Arctic Shift	Access and policy conditions	Keep adapter disabled
Sentiment	Irony and financial context	Manual sample review
Scale	14,773 price rows initially	Measure actual social volume

Optional later: images, OCR, embeddings and grounded summaries

Sources and credits

Project evidence

Team CMC CSVs and EDA notebook
Proposal and architecture, 14 Oct 2026

Source and database documentation

MemeChain on Zenodo, FxTwitter, Arctic Shift
PostgreSQL, MongoDB and Reddit policy

Visual direction and images

shadcn/ui and Vercel Geist
Atsuko Sato / KAI-YOU, GetTrumpMemes, CMC, PostgreSQL, MongoDB

Full URLs and image attribution appear in each slide’s speaker notes and sources.md.